DEGREES OF FIELD EXTENSIONS ARE MULTIPLICATIVE IN TOWERS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: This is the workhorse behind every degree count in the chapter, including the multiplicativity steps invoked by the embedding-count lemmas.

Editorial note. The label lemma-multiplicativity-degrees names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Vector space bases: an E-basis of F and a k-basis of E; k-linear independence in E

Lemma 0.1 (Multiplicativity)**.** *Suppose given a tower of fields $F/E/k$. Then*

$$[F : k] = [F : E][E : k]$$

Proof. Let $\alpha_1, \ldots, \alpha_n \in F$ be an E-basis for F. Let $\beta_1, \ldots, \beta_m \in E$ be a k-basis for E. Then the claim is that the set of products $\{\alpha_i\beta_j, 1 \leq i \leq n, 1 \leq j \leq m\}$ is a k-basis for F. Indeed, let us check first that they span F over k.

By assumption, the $\{\alpha_i\}$ span F over E. So if $f \in F$, there are $a_i \in E$ with

$$f = \sum\nolimits_i a_i\alpha_i,$$

and, for each i, we can write $a_i = \sum b_{ij}\beta_j$ for some $b_{ij} \in k$. Putting these together, we find

$$f = \sum\nolimits_{i,j} b_{ij}\alpha_i\beta_j,$$

proving that the $\{\alpha_i\beta_j\}$ span F over k.

Suppose now that there existed a nontrivial relation

$$\sum\nolimits_{i,j} c_{ij}\alpha_i\beta_j = 0$$

for the $c_{ij} \in k$. In that case, we would have

$$\sum\nolimits_i \alpha_i \left(\sum\nolimits_j c_{ij}\beta_j\right) = 0,$$

and the inner terms lie in E as the β_j do. Now E-linear independence of the $\{\alpha_i\}$ shows that the inner sums are all zero. Then k-linear independence of the $\{\beta_j\}$ shows that the c_{ij} all vanish. □

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